# Brains and Models

# 1. Structural lens: synapses and weights

A simple model assigns one independent scalar weight to each synapse.

## $P \approx S = N \times k$

N = neurons in the chosen region  k = mean incoming synapses per neuron

**Regional scope.** Use N and k from the same region. For a whole-brain estimate, sum N × k across regions and count each synapse once.

**Human example.** About 86 billion neurons in the brain, about 16 billion in the cerebral cortex. One neocortical study estimated $1.64 \times 10^{14}$ synapses. 164 Trillion

**Limit.** This approximation counts connections. Weight sharing, synaptic dynamics and neuron properties change the relationship to model parameters.

Azevedo et al., 2009; Tang et al., 2001. Full citations in speaker notes.

# 2. Storage lens: bits per synapse and per weight

**Evidence.** Bartol et al. inferred at least 26 distinguishable strength levels (about 4.7 bits) in sampled rat hippocampal synapses.

This is a lower-bound estimate under that study’s assumptions. Applying it to the human brain requires an explicit extrapolation.

## Equivalent weight slots ≈ S × b / q

S = synapses  b = assumed bits per synapse  q = stored bits per weight

**What it measures.** A nominal storage comparison at a stated precision.

**Limits.** Stored bit width differs from learned information. Correlations and unequal state probabilities affect capacity. Other biological states add detail.

Bartol et al., 2015. Full citation and assumptions in speaker notes.

# 3. Graph lens: a description of the wiring

A simple edge list stores both endpoints and a strength for each connection.

## Bits per edge ≈ $2\lceil \log_2 N \rceil + b$

With N = 86 billion neurons: 37 bits per endpoint, or 74 address bits.

If b = 4.7, the idealized total is 78.7 bits per edge.
That equals about 4.9 storage words of 16 bits each.

**Encoding matters.** Adjacency lists, spatial structure and compression can reduce this description length. It is a property of the chosen encoding.

**Meaning.** These are storage-word equivalents. Dividing by 16 does not turn connection addresses into additional learned parameters.

Address arithmetic is illustrative. Inputs: Azevedo et al., 2009; Bartol et al., 2015.

# 4. Dynamics lens: reuse of connections over time

Unrolling a recurrent model reuses the same weights at successive time steps.

## Connection evaluations ≈ a × S × T

a = fraction of connections evaluated per step  T = number of steps

**Illustrative example.** $S = 1.6 \times 10^{14}$, $T = 100$ and $a = 1$ give $1.6 \times 10^{16}$ connection evaluations.

**Unique weights.** The model still has $1.6 \times 10^{14}$ synaptic weights under the one-weight-per-synapse assumption.

**Caveat.** T and a depend on the model. A synaptic event has no universal conversion to FLOPs, and biological activity is asynchronous.

Recurrent weight reuse: PyTorch RNN documentation. Biological scale: Tang et al., 2001.

# 5. Regional lens: a brain area and a model stage

A regional comparison keeps the biological scope closer to the task.

**Scope.** Choose a region, such as primary visual cortex (V1),
and define which local and long-range inputs count.

**Connections.** Estimate $S_r = N_r \times k_r$ using compatible regional data.
An ANN with untied scalar weights can match that connection count.

**Function.** Test predictions of neural responses or behavior.
Matching counts alone does not establish comparable capability.

**Biological detail.** A neuron can have nonlinear dendrites and changing internal state. Feedback and plasticity affect the comparison.

Azevedo et al., 2009; Beniaguev et al., 2021; PyTorch Conv2d documentation.

# Worked example: human neocortical scale

Use $S \approx 1.6 \times 10^{14}$ synapses, rounded from a neocortical study.
For the storage calculation only, assume b = 4.7 bits per synapse.

| Comparison | Calculation | Result |
|---|---|---|
| Synaptic weights | $P \approx S$ | 160 trillion weights |
| 16-bit equivalents | $S \times 4.7 / 16$ | 47 trillion slots |
| 8-bit equivalents | $S \times 4.7 / 8$ | 94 trillion slots |

The first row counts independent weights in an idealized model.
The other rows compare nominal storage under an extrapolated assumption.

Tang et al., 2001 (human neocortex); Bartol et al., 2015 (rat hippocampus).

# Comparisons across mammals

**1. Define the population.** Use the species, brain region and age group relevant to the question. Record sources and uncertainty.

**2. Count consistently.** Use regional N and incoming k, or measured synapse density × regional volume. Sum regions only with compatible boundaries.

**3. Choose the quantity.** Report connections, storage bits, wiring description or connection evaluations. Each quantity has different assumptions.

**4. Check sensitivity.** Vary uncertain inputs. State weight sharing, precision and time horizon before interpreting the result.

For example, about 71 million neurons is a published whole-mouse-brain estimate. A cortical synapse-per-neuron value needs a cortical neuron count.

Herculano-Houzel et al., 2006. Full citation in speaker notes.

# Summary: five ways to compare brains and models

| Lens | Quantity | What it tells us |
|---|---|---|
| Structure | Synapses S | Count of connections |
| Storage | S × assumed bits per synapse | Nominal storage budget |
| Wiring | Address bits + strength bits | Description length |
| Dynamics | Connection evaluations | Work over a defined interval |
| Region | A brain area and model stage | A functional hypothesis to test |

Parameter count, storage and computation answer different questions.
The biological scope and modeling assumptions determine the comparison.

# Summary: the human numbers and their limits

**≈86 billion neurons.** A whole-brain estimate, with about 16 billion in the cerebral cortex. Regional differences matter.

**≈160 trillion synapses.** A rounded estimate for the human neocortex from one study, rather than a whole-brain total.

**≈47 trillion 16-bit slots.** A conditional storage equivalent if every synapse contributes 4.7 bits. That assumption extrapolates from rat hippocampus.

**These estimates describe different quantities and populations. They do not establish a human-equivalent AI model size.**

Azevedo et al., 2009; Tang et al., 2001; Bartol et al., 2015.

# Summary: what makes a useful comparison

**A clear question.** The relevant brain region and task define the scope.

**Explicit units.** Connections, bits and operations remain separate quantities.

**Visible assumptions.** Precision, weight sharing and time determine how a biological count maps to a model.

**Functional evidence.** Neural or behavioral tests are needed to evaluate what a model captures about the brain.

**There is no single conversion from synapses to AI capability.**